Frequency of bacteria causing urinary tract infections and their antimicrobial resistance patterns among pediatric patients in Western Iran from 2007-2009.

PMID: 25316131 | Indexed in PubMed

Urinary Tract infections (UTIs) are among the most common infections in infants and neonates. The aim of the current study was to evaluate the frequency of bacteria causing UTI and their relevant drug resistance patterns among infants and neonates hospitalized in Ilam province, Western Iran during 2007-2009. A total of 220 cases of UTI were enrolled in this cross-sectional retrospective study. A standard checklist was used for demographic and clinical data to be collected from their health records. Data was then analyzed using SPSS version 16.0. More than two-thirds (64.8%) of the cases were female. E. coli (44.5%), Klebsiella spp., (18.6%), Enterobacter spp., (15%) and Staphylococcus spp. (12.7%) were the most common microorganisms isolated from UTIs, respectively. High rates of resistance to tetracycline, ampicillin, and nalidixic acid were observed among these isolates. Similar to other studies, E. coli was the most common bacteria causing UTI and showed a high rate of resistance against most of the antimicrobial agents. Determining the antimicrobial sensitivity can be helpful for physicians in choosing an appropriate treatment for patients suffering from UTI, and also to reduce the complications related to serious UTI.

Outbreak of drug resistance Escherichia coli phylogenetic F group associated urinary tract infection.

PMID: 37124864 | Indexed in PubMed

Urinary tract infections are one of the most commonly associated human infectious diseases caused by the bacteria Escherichia coli. Escherichia coli is described as having a large number of virulence genes that enable drug resistance, which is a cause for great concern. Monitoring of antimicrobial susceptibility is critical to determining the scope of the problem and selecting appropriate antimicrobial drugs. The current study aimed to identify the distribution of uropathogenic E. coli (UPEC) based on genetic profiles and to determine resistance patterns among isolates. This study employed biological correlations to study the patterns of antibiotic resistance and the distribution of phylogenetic groups of 118 isolates of E. coli and the relationship between them, which were isolated from three hospitals in Baghdad, Iraq. The results of phylogenetic analysis showed that phylogroup F was the most common group among E. coli isolates (37.3%), followed by phylogroups C (20.3%), B2 (15.3%), E (14.4%), UP (4.2%), A and D (3.4%), and B1 (1.7%). The majority of antibiotic resistance patterns were related to penicillin groups (80.5%) and the least to the sulfonamide groups (67.0%). 51.7%, 42.4%, and 1.7% of isolates were Extensive Drug Resistance (XDR), Multi-Drug Resistance (MDR), and Pan Drug Resistance (PDR), respectively. Antibiotic resistance was most commonly detected in group F (35.6%). Our observations revealed that the dominant phylogroup F had the highest prevalence of multi-drug resistance and extensive drug resistance among E. coli isolates. The newly identified phylogroups C, E, and F account for about 72.0% of the E. coli isolates. Such investigations should be conducted in other localities as well, in order to gain a better understanding of the pattern of antibiotic resistance patterns and the frequency of distinct phylogenetic groups.

[New, successful treatment of urinary tract infection caused by Pseudomonas

aeruginosa].

PMID: 16898513 | Indexed in PubMed

Urinary tract infection (UTI) is common in pediatric practice and an important cause of morbidity and mortality in children. Escherichia coli remains the predominant uropathogen (80%) isolated in acute community-acquired uncomplicated infections, followed by Staphylococcus saprophyticus (10% to 15%) and Pseudomonas aeruginosa (9%) The pathogens traditionally associated with UTI are changing many of their features, particularly because of antimicrobial resistance. Reinfections and relapses of urinary tract infections caused by PA are very frequent. The aim of the study was to evaluate the efficacy of combined clarithromycine and ceftazidime in terms of eradication of PA infection. We analyzed 20 out of 264 children with UTI where PA infection was confirmed with urine culture. Those children were treated for at least 14 days with the protocol used for PA infection in patients with mucoviscidosis. Short-term eradication was achieved in all patients. Long-term study revealed relapse in 25% of children, all with serious congenital malformations. 75% of children were treated with success. No side effects were observed. Conclusion. We conclude that an empirical combination treatment of clarithromycine and ceftazidime is appropriate and effective in children with UTI caused by PA. This therapy was clinically efficacious, well tolerated, and cost effective, and should prevent unnecessary development of antimicrobial resistance.

Population dynamics and niche distribution of uropathogenic Escherichia coli during acute and chronic urinary tract infection.

PMID: 21807904 | Indexed in PubMed

Urinary tract infections (UTIs) have complex dynamics, with uropathogenic Escherichia coli (UPEC), the major causative agent, capable of colonization from the urethra to the kidneys in both extracellular and intracellular niches while also producing chronic persistent infections and frequent recurrent disease. In mouse and human bladders, UPEC invades the superficial epithelium, and some bacteria enter the cytoplasm to rapidly replicate into intracellular bacterial communities (IBCs) comprised of ~10^4 bacteria each. Through IBC formation, UPEC expands in numbers while subverting aspects of the innate immune response. Within 12 h of murine bladder infection, half of the bacteria are intracellular, with 3 to 700 IBCs formed. Using mixed infections with green fluorescent protein (GFP) and wild-type (WT) UPEC, we discovered that each IBC is clonally derived from a single bacterium. Genetically tagged UPEC and a multiplex PCR assay were employed to investigate the distribution of UPEC throughout urinary tract niches over time. In the first 24 h postinfection (hpi), the fraction of tags dramatically decreased in the bladder and kidney, while the number of CFU increased. The percentage of tags detected at 6 hpi correlated to the number of IBCs produced, which closely matched a calculated multinomial distribution based on IBC clonality. The fraction of tags remaining thereafter depended on UTI outcome, which ranged from resolution of infection with or without quiescent intracellular reservoirs (QIRs) to the development of chronic cystitis as defined by persistent bacteriuria. Significantly more tags remained in mice that developed chronic cystitis, arguing that during the acute stages of infection, a higher number of IBCs precedes chronic cystitis than precedes QIR formation.

Antimicrobial Agents and Urinary Tract Infections.

PMID: 31218955 | Indexed in PubMed

Urinary Tract Infections (UTIs); second-ranking infectious diseases are regarded as a significant global health care problem. The UTIs annually cost tens of millions of dollars for governments worldwide. The main reason behind these costs is incorrect or indefinite treatment. There are a wide range of gram-negative and grampositive bacteria which may cause UTIs in males and females, children and adults. Among gram-negative bacteria, some members of Enterobacteriaceae such as Escherichia coli (E.coli) strains have significant contribution in UTIs. Uropathogenic E.coli (UPEC) strains are recognized as typical bacterial agents for UTIs. Thus, sharp and accurate diagnostic tools are needed for detection and identification of the microbial causative agents of UTIs. In parallel with the utilization of suitable diagnostic methods-to reduce the number of UTIs, effective and definite treatment procedures are needed. Therefore, the prescription of accurate, specific and effective antibiotics and drugs may lead to a definite treatment. However, there are many cases related to UTIs which can be relapsed. Due to a diversity of opportunistic and pathogenic causative microbial agents of UTIs, the treatment procedures should be achieved by the related antimicrobial agents. In this review, common and effective antimicrobial agents which are often prescribed for UTIs caused by UPEC will be discussed. Moreover, we will have a sharp look at their (antimicrobials) molecular treatment mechanisms.

Antibody-Based PET Imaging of Misfolded Superoxide Dismutase 1 in an Amyotrophic Lateral Sclerosis Mouse Model.

PMID: 39667814 | Indexed in PubMed

Amyotrophic lateral sclerosis (ALS) is a rare neurodegenerative disease characterized by motor neuron loss in the motor cortex, brain stem, and spinal cord. Mutations in the superoxide dismutase 1 (SOD1) gene, resulting in misfolding of its protein product, are a common cause of ALS. Currently, there is no approved ALS diagnostic tool. Here, we present the development of a PET radiotracer, [89Zr]Zr-desferoxamine (DFO)-α-miSOD1, targeting selectively misfolded SOD1 (misSOD1). Methods: DFO-α-miSOD1 was prepared by conjugating α-miSOD1 antibody with DFO and labeled with 89Zr. A longitudinal imaging study was performed to identify the optimal mouse age and time after administration of [89Zr]Zr-DFO-α-miSOD1 for the detection of misSOD1 aggregation in transgenic mice overexpressing misSOD1 and in wild-type mice. Subsets of mice were either coinjected with an excess of α-miSOD1 or imaged with deglycosylated [89Zr]Zr-DFO-α-miSOD1 to assess target specificity. The internal radiation dose for [89Zr]Zr-DFO-α-miSOD1 was estimated by extrapolating data from mouse biodistribution experiments. Results: Imaging with [89Zr]Zr-DFO-α-miSOD1 was optimal in 136-d-old transgenic mice on day 10 after administration. Significant accumulation of [89Zr]Zr-DFO-α-miSOD1 was detected in the spinal cord and cartilage of ALS transgenic mice compared with the wild-type mice ($P = 0.01$). The radiotracer accumulation is selective and blockable with an excess of α-miSOD1. Deglycosylated [89Zr]Zr-DFO-α-miSOD1 results in high-contrast detection of misSOD1 but is prone to aggregation. The dosimetry for [89Zr]Zr-DFO-α-miSOD1 is comparable to that for other 89Zr-based tracers currently used in humans. Conclusion: This work thus establishes that [89Zr]Zr-DFO-α-miSOD1 PET can detect misSOD1 in transgenic mice, paving the way for application in early diagnosis of ALS and therapeutic monitoring.

Pathogenesis of Neurodegenerative Diseases and the Protective Role of Natural

Bioactive Components.

PMID: 37186678 | Indexed in PubMed

Neurodegenerative diseases are a serious problem throughout the world. There are several causes of neurodegenerative diseases; these include genetic predisposition, accumulation of misfolded proteins, oxidative stress, neuroinflammation, and excitotoxicity. Oxidative stress increases the production of reactive oxygen species (ROS) that advance lipid peroxidation, DNA damage, and neuroinflammation. The cellular antioxidant system (superoxide dismutase, catalase, peroxidase, and reduced glutathione) plays a crucial role in scavenging free radicals. An imbalance in the defensive actions of antioxidants and overproduction of ROS intensify neurodegeneration. The formation of misfolded proteins, glutamate toxicity, oxidative stress, and cytokine imbalance promote the pathogenesis of Alzheimer's disease, Parkinson's disease, Huntington's disease, and amyotrophic lateral sclerosis. Antioxidants are now attractive molecules to fight against neurodegeneration. Certain vitamins (A, E, C) and polyphenolic compounds (flavonoids) show excellent antioxidant properties. Diet is the major source of antioxidants. However, diet medicinal herbs are also rich sources of numerous flavonoids. Antioxidants prevent ROS-mediated neuronal degeneration in post-oxidative stress conditions. The present review is focused on the pathogenesis of neurodegenerative diseases and the protective role of antioxidants. KEY TEACHING POINTSThis review shows that multiple factors are directly or indirectly associated with the pathogenesis of neurodegenerative diseases.Failure to cellular antioxidant capacity increases oxidative stress that intensifies neuroinflammation and disease progression.Different vitamins, carotenoids, and flavonoids, having antioxidant capacity, can be considered protective agents.

Human prion disease: molecular pathogenesis, and possible therapeutic targets and strategies.

PMID: 37334903 | Indexed in PubMed

Human prion diseases are heterogeneous, and often rapidly progressive, transmissible neurodegenerative disorders associated with misfolded prion protein (PrP) aggregation and self-propagation. Despite their rarity, prion diseases comprise a broad spectrum of phenotypic variants determined at the molecular level by different conformers of misfolded PrP and host genotype variability. Moreover, they uniquely occur in idiopathic, genetically determined, and acquired forms with distinct etiologies. This review provides an up-to-date overview of potential therapeutic targets in prion diseases and the main results obtained in cell and animal models and human trials. The open issues and challenges associated with developing effective therapies and informative clinical trials are also discussed. Currently tested therapeutic strategies target the cellular PrP to prevent the formation of misfolded PrP or to favor its elimination. Among them, passive immunization and gene therapy with antisense oligonucleotides against prion protein mRNA are the most promising. However, the disease's rarity, heterogeneity, and rapid progression profoundly frustrate the successful undertaking of well-powered therapeutic trials and patient identification in the asymptomatic or early stage before the development of significant brain damage. Thus, the most promising therapeutic goal to date is preventing or delaying phenoconversion in carriers of pathogenic mutations by lowering prion protein expression.

Neurodegenerative disorders of protein aggregation.

PMID: 12605877 | Indexed in PubMed

In recent years, it has become increasingly clear that many neurodegenerative diseases involve aggregation and deposition of misfolded proteins such as amyloid beta, tau, alpha-synuclein and polyglutamine containing proteins. This abnormal deposition of misfolded proteins produce malfunctioning of a distinctive set of neurons. It may also induce oxidative and endoplasmic reticulum stress and proteosomal and mitochondrial dysfunction that ultimately leads to neuronal death. While hereditary forms of disorders are caused by genetic mutations, many sporadic cases are likely to be due to genetic and environmental factors. These disorders are progressive in nature. Therefore, treatment is difficult. However, for some diseases, a growing number of treatment options such as drugs, antioxidants, cell transplantation, surgery, rehabilitation procedures and preimplantation diagnosis is available. It should be noted that many of these treatments produce unacceptable risks or adverse effects and they are of only minimal benefit for patients. In future, an understanding of the causes of protein aggregation and genetic and environmental susceptibility factors of a specific individual (or specific individual determinants) may provide a better opportunity for an effective therapeutic intervention.

Molecular Chaperones as Therapeutic Target: Hallmark of Neurodegenerative Disorders.

PMID: 38127187 | Indexed in PubMed

Misfolded and aggregated proteins build up in neurodegenerative illnesses, which causes neuronal dysfunction and ultimately neuronal death. In the last few years, there has been a significant upsurge in the level of interest towards the function of molecular chaperones in the control of misfolding and aggregation. The crucial molecular chaperones implicated in neurodegenerative illnesses are covered in this review article, along with a variety of their different methods of action. By aiding in protein folding, avoiding misfolding, and enabling protein breakdown, molecular chaperones serve critical roles in preserving protein homeostasis. By aiding in protein folding, avoiding misfolding, and enabling protein breakdown, molecular chaperones have integral roles in preserving regulation of protein balance. It has been demonstrated that aging, a significant risk factor for neurological disorders, affects how molecular chaperones function. The aggregation of misfolded proteins and the development of neurodegeneration may be facilitated by the aging-related reduction in chaperone activity. Molecular chaperones have also been linked to the pathophysiology of several instances of neuron withering illnesses, enumerating as Parkinson's disease, Huntington's disease, and Alzheimer's disease. Molecular chaperones have become potential therapy targets concerning with the prevention and therapeutic approach for brain disorders due to their crucial function in protein homeostasis and their connection to neurodegenerative illnesses. Protein homeostasis can be restored, and illness progression can be slowed down by methods that increase chaperone function or modify their expression. This review emphasizes the importance of molecular chaperones in the context of neuron withering disorders and their potential as therapeutic targets for brain disorders.